

- 3 A mass P, 80 g, is attached to the free end of a horizontal spring on a smooth surface. The spring-mass system is set into simple harmonic motion by pulling P to the right of the equilibrium position and is released from rest as shown in Fig. 3.1.

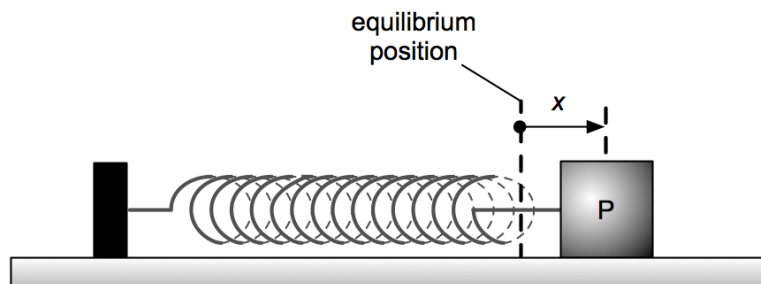


Fig. 3.1

If the air resistance on P is negligible, the variation of the velocity v of P with displacement x is shown in Fig. 3.2. Vectors to the right are taken to be positive.

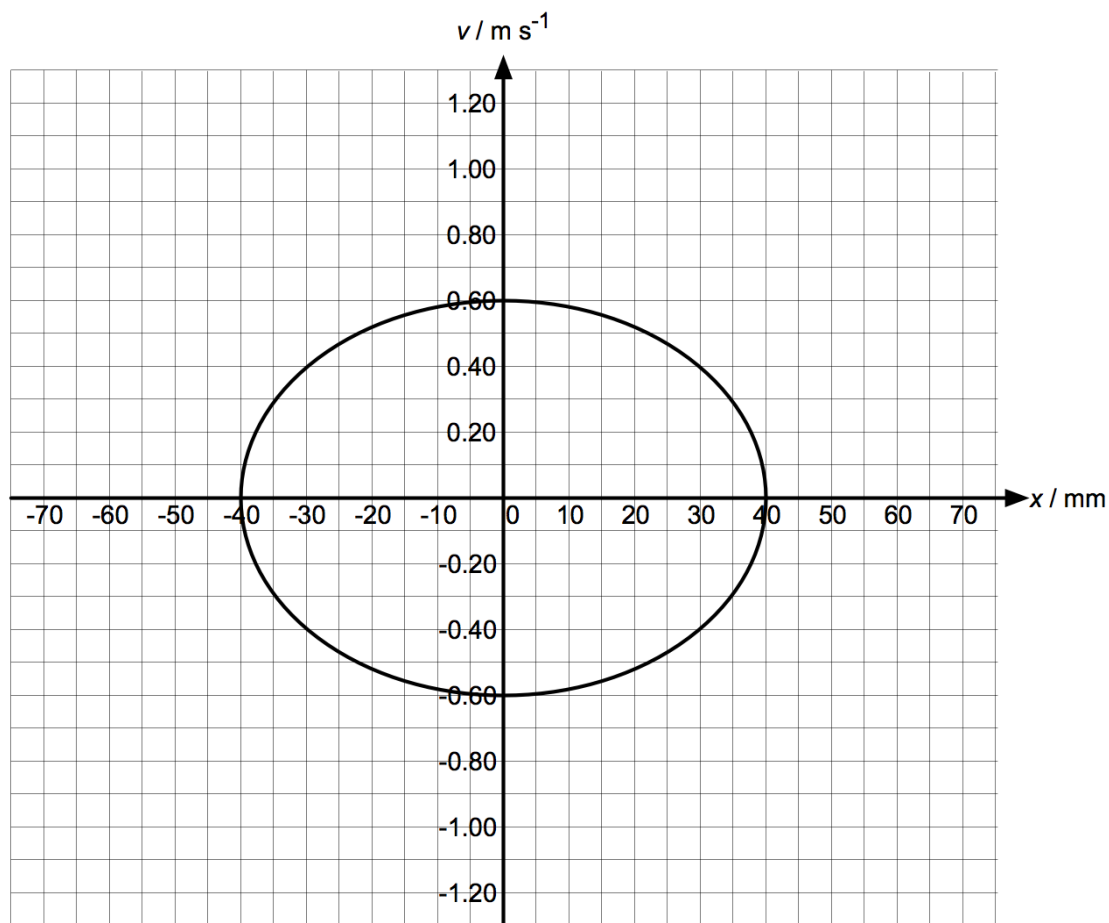


Fig. 3.2

- (a) For the motion of P, determine
- (i) the amplitude,

40 mm

amplitude = mm [1]

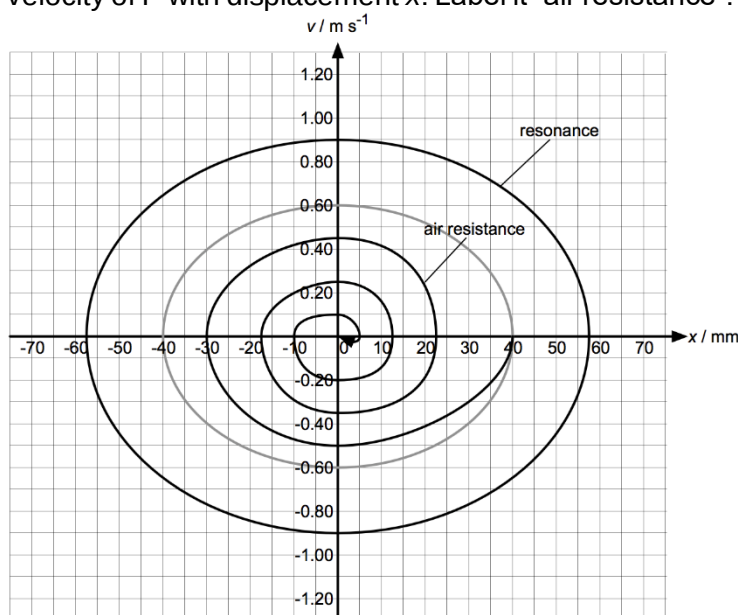
- (ii) the frequency.

$$v_{\max} = \omega x_0 \Rightarrow 0.60 = (2\pi f)(0.040)$$

$$f = 2.4 \text{ Hz}$$

frequency = Hz [2]

- (b) If the air resistance on P is not negligible, sketch on Fig. 3.2 the variation of the velocity of P with displacement x. Label it “air resistance”. [3]



Must start at (40, 0) [1] (displaced to the right, right is positive)
 Clockwise spiral [1] (initially move to left so starting velocity is negative)
 Spiral towards origin. [1]

- (c) A periodic force is now exerted on the spring-mass system. When the periodic force is at a certain frequency, P is in resonance.

- (i) Given that the total energy of the spring-mass system at steady state is doubled. Determine the new maximum speed of P.

$$E_T = \frac{1}{2}mv_0^2 \Rightarrow E_T \propto v_0^2 \Rightarrow \frac{E'_T}{E_T} = \frac{v_0'^2}{v_0^2} \Rightarrow \frac{2E_T}{E_T} = \frac{v_0'^2}{0.6^2}$$

$$v_0' = 0.85 \text{ m s}^{-1}$$

maximum speed of P = m s⁻¹ [2]

- (ii) On Fig. 3.2, sketch the variation of the velocity of P, at resonance, with displacement x. Label it “resonance”. [2]

At resonance, frequency of P is near to its natural frequency of 2.4 Hz.

$$v_0 = \omega x_0 \Rightarrow v_0 \propto x_0 \Rightarrow x_0' = \sqrt{2}x_0 = 57 \text{ mm}.$$

Elliptical shape with maximum speed at 0.85 m s⁻¹ and amplitude at 57 mm.

4 (a) Define the electric potential at a point in an electric field